DSBDA Assignment No.13

Spark installation:

```
sujal@Sujal-PC09: ~
sujal@Sujal-PC09:~$ sudo apt update
sudo apt install openjdk-11-jdk -y
[sudo] password for sujal:
Get:1 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Get:4 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Hit:3 https://scala.jfrog.io/artifactory/debian all InRelease
Get:5 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [741 kB]
Hit:6 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Get:7 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [991 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [828 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1052 kB]
Fetched 3865 kB in 5s (812 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
3 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
openjdk-11-jdk is already the newest version (11.0.26+4-1ubuntu1~24.04).
0 upgraded, 0 newly installed, 0 to remove and 3 not upgraded.
sujal@Sujal-PC09:~$ java -version
openjdk version "11.0.26" 2025-01-21
OpenJDK Runtime Environment (build 11.0.26+4-post-Ubuntu-1ubuntu124.04)
OpenJDK 64-Bit Server VM (build 11.0.26+4-post-Ubuntu-1ubuntu124.04, mixed mode, sharing)
sujal@Sujal-PC09:~$ cd ~
sujal@Sujal-PC09:~$ wget https://downloads.apache.org/spark/spark-3.5.0/spark-3.5.0-bin-hadoop3.tgz
--2025-04-09 04:16:28--  https://downloads.apache.org/spark/spark-3.5.0/spark-3.5.0-bin-hadoop3.tgz
Resolving downloads.apache.org (downloads.apache.org)... 88.99.208.237, 135.181.214.104, 2a01:4f9:3a:2c57::2, ...
Connecting to downloads.apache.org (downloads.apache.org)|88.99.208.237|:443... connected.
HTTP request sent, awaiting response... 404 Not Found
2025-04-09 04:16:32 ERROR 404: Not Found.

sujal@Sujal-PC09:~$ tar -xvzf spark-3.5.0-bin-hadoop3.tgz
tar (child): spark-3.5.0-bin-hadoop3.tgz: Cannot open: No such file or directory
tar (child): Error is not recoverable: exiting now
tar: Child returned status 2
tar: Error is not recoverable: exiting now
sujal@Sujal-PC09:~$ mv spark-3.5.0-bin-hadoop3 spark
mv: cannot stat 'spark-3.5.0-bin-hadoop3': No such file or directory
```

```
sujal@Sujal-PC09: ~

sujal@Sujal-PC09:~$ wget https://downloads.apache.org/spark/spark-3.5.5/spark-3.5.5-bin-hadoop3.tgz
--2025-04-09 04:18:36--  https://downloads.apache.org/spark/spark-3.5.5/spark-3.5.5-bin-hadoop3.tgz
Resolving downloads.apache.org (downloads.apache.org)... 135.181.214.104, 88.99.208.237, 2a01:4f9:3a:2c57::2, ...
Connecting to downloads.apache.org (downloads.apache.org)|135.181.214.104|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 400724056 (382M) [application/x-gzip]
Saving to: ‘spark-3.5.5-bin-hadoop3.tgz’

spark-3.5.5-bin-ha 100%[===============>] 382.16M  4.71MB/s    in 1m 44s

2025-04-09 04:20:21 (3.69 MB/s) - ‘spark-3.5.5-bin-hadoop3.tgz’ saved [400724056/400724056]

sujal@Sujal-PC09:~$ tar -xvzf spark-3.5.5-bin-hadoop3.tgz
spark-3.5.5-bin-hadoop3/
spark-3.5.5-bin-hadoop3/jars/
spark-3.5.5-bin-hadoop3/jars/HikariCP-2.5.1.jar
spark-3.5.5-bin-hadoop3/jars/JLargeArrays-1.5.jar
spark-3.5.5-bin-hadoop3/jars/JTransforms-3.1.jar
spark-3.5.5-bin-hadoop3/jars/RoaringBitmap-0.9.45.jar
spark-3.5.5-bin-hadoop3/jars/ST4-4.0.4.jar
spark-3.5.5-bin-hadoop3/jars/activation-1.1.1.jar
spark-3.5.5-bin-hadoop3/jars/aircompressor-0.27.jar
spark-3.5.5-bin-hadoop3/jars/algebra_2.12-2.0.1.jar
spark-3.5.5-bin-hadoop3/jars/annotations-17.0.0.jar
spark-3.5.5-bin-hadoop3/jars/antlr-runtime-3.5.2.jar
spark-3.5.5-bin-hadoop3/jars/antlr4-runtime-4.9.3.jar
spark-3.5.5-bin-hadoop3/jars/aopalliance-repackaged-2.6.1.jar
spark-3.5.5-bin-hadoop3/jars/arpack-3.0.3.jar
spark-3.5.5-bin-hadoop3/jars/arpack_combined_all-0.1.jar
spark-3.5.5-bin-hadoop3/jars/arrow-format-12.0.1.jar
spark-3.5.5-bin-hadoop3/jars/arrow-memory-core-12.0.1.jar
spark-3.5.5-bin-hadoop3/jars/arrow-memory-netty-12.0.1.jar
spark-3.5.5-bin-hadoop3/jars/arrow-vector-12.0.1.jar
spark-3.5.5-bin-hadoop3/jars/audience-annotations-0.5.0.jar
spark-3.5.5-bin-hadoop3/jars/avro-1.11.4.jar
spark-3.5.5-bin-hadoop3/jars/avro-ipc-1.11.4.jar
spark-3.5.5-bin-hadoop3/jars/avro-mapred-1.11.4.jar
spark-3.5.5-bin-hadoop3/jars/blas-3.0.3.jar
spark-3.5.5-bin-hadoop3/jars/bonecp-0.8.0.RELEASE.jar
spark-3.5.5-bin-hadoop3/jars/breeze-macros_2.12-2.1.0.jar
```

```
spark-3.5.5-bin-hadoop3/R/lib/SparkR/profile/shell.R
spark-3.5.5-bin-hadoop3/R/lib/SparkR/tests/
spark-3.5.5-bin-hadoop3/R/lib/SparkR/tests/testthat/
spark-3.5.5-bin-hadoop3/R/lib/SparkR/tests/testthat/test_basic.R
spark-3.5.5-bin-hadoop3/R/lib/SparkR/worker/
spark-3.5.5-bin-hadoop3/R/lib/SparkR/worker/daemon.R
spark-3.5.5-bin-hadoop3/R/lib/SparkR/worker/worker.R
spark-3.5.5-bin-hadoop3/R/lib/SparkR/help/
spark-3.5.5-bin-hadoop3/R/lib/SparkR/help/paths.rds
spark-3.5.5-bin-hadoop3/R/lib/SparkR/help/SparkR.rdb
spark-3.5.5-bin-hadoop3/R/lib/SparkR/help/SparkR.rdx
spark-3.5.5-bin-hadoop3/R/lib/SparkR/help/AnIndex
spark-3.5.5-bin-hadoop3/R/lib/SparkR/help/aliases.rds
spark-3.5.5-bin-hadoop3/R/lib/SparkR/html/
spark-3.5.5-bin-hadoop3/R/lib/SparkR/html/00Index.html
spark-3.5.5-bin-hadoop3/R/lib/SparkR/html/R.css
spark-3.5.5-bin-hadoop3/R/lib/SparkR/INDEX
spark-3.5.5-bin-hadoop3/R/lib/sparkr.zip
sujal@Sujal-PC09:~$ mv spark-3.5.5-bin-hadoop3 spark
sujal@Sujal-PC09:~$ nano ~/.bashrc
sujal@Sujal-PC09:~$ source ~/.bashrc
sujal@Sujal-PC09:~$ spark-shell
25/04/09 04:21:02 WARN Utils: Your hostname, Sujal-PC09 resolves to a loopback address: 127.0.1.1; using 10.255.255.254 instead (on interface lo)
25/04/09 04:21:02 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another address
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
25/04/09 04:21:06 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Spark context Web UI available at http://10.255.255.254:4040
Spark context available as 'sc' (master = local[*], app id = local-1744172466775).
Spark session available as 'spark'.
Welcome to
      ____              __
     / __/__  ___ _____/ /__
    _\ \/ _ \/ _ `/ __/  '_/
   /___/ .__/\_,_/_/ /_/\_\   version 3.5.5
      /_/

Using Scala version 2.12.18 (OpenJDK 64-Bit Server VM, Java 11.0.26)
Type in expressions to have them evaluated.
Type :help for more information.
```

Q. Find given number is Positive/Negative using 'If' Statement

```
scala> :q
sujal@Sujal-PC09:~$ nano CheckNum.scala
sujal@Sujal-PC09:~$ scalac CheckNum.scala
scala CheckNum
Enter number: 14
Number is POSITIVE
sujal@Sujal-PC09:~$ cat CheckNum.scala
object CheckNum {
  def main(args: Array[String]): Unit = {
    print("Enter number: ")
    val num = scala.io.StdIn.readInt()
    if (num >= 0)
      println("Number is POSITIVE")
    else
      println("Number is NEGATIVE")
  }
}
sujal@Sujal-PC09:~$
```